

The power of nonverbal behavior: Penalty-takers' body language influences impression formation and anticipation performance in goalkeepers in a simulated soccer penalty task



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ABSTRACT

Previous research showed that penalty-takers' body language affects the impressions goalkeepers form about them. Furthermore, it is often assumed that body language influences goalkeepers' performance. However, as yet, there is no empirical evidence that indicates whether this latter effect exists. The present research was aimed at (1) replicating and offering more insight in the initial impression formation effects (Experiment 1-3) and (2) providing evidence that signals of dominance and submissiveness affect participants' anticipation performance within a simulated soccer penalty task (Experiment 3). We report three experiments demonstrating that participants form more positive impressions and have less confidence in saving penalty kicks from dominant penalty-takers (or scoring against dominant goalkeepers, Experiment 1) than submissive penalty-takers (goalkeepers). However, we did not find evidence that participants' automatic associations with dominant and submissive players underlie these findings (Experiment 2). Finally, we demonstrate that anticipation of kick direction is influenced by penalty-takers' body language (Experiment 3). Participants performed worse in the simulated soccer penalty task against dominant than submissive penalty-takers. No mediation of impression formation was found.

Since their introduction in 1978, penalty shootouts have decided 30 FIFA Soccer World Cup matches and two world cup trophies (i.e., in 1994 and 2006). Being ultimately decisive, penalty shootouts may be regarded as the pinnacle of high-pressure performance (Wood, Jordet, & Wilson, 2015), as players engage in a 'mental game' of influencing their opponents' emotions and behavior in an attempt to gain competitive advantage (e.g., Jordet & Elferink-Gemser, 2012). Investigating the social dynamics that occur between penalty-takers and goalkeepers, recent research has shown that pre-performance nonverbal behavior (colloquially referred to as 'body language') that is displayed by penalty-takers influences impression formation processes in observing goalkeepers and alters their confidence in being able to stop the upcoming penalty kick (e.g., Furley, Dicks, & Memmert, 2012). Building on this finding, the goal of the present research was to further understanding of how body language influences opponents' impressions and to examine how it may affect performance in a simulated soccer penalty situation.

A large body of psychological research has shown that people are

constantly making judgments and forming impressions of other people based on observable cues (reviewed in Freeman & Ambady, 2011). Pertinent to the present research, cues that have been shown to be of considerable ecological importance, and of particular importance in sports (Furley & Schweizer, 2016), are dominance cues (Furley, Schnuerch, & Gibbons, 2017; Mazur, 2005; Rule, Adams, Ambady, & Freeman, 2012; Zebrowitz & Collins, 1997). To investigate the impact of dominance cues in soccer penalty situations, Furley, Dicks, and Memmert (2012; Experiment 1) had participants watch videos of penalty-takers who either displayed *dominant body language* (i.e., erect posture, shoulders back, chest out, head straight) or *submissive body language* (i.e., slouched posture, shoulders forward, chest in, head down) as they prepared for a penalty kick, and asked them to report back their impressions of the observed players. Confirming earlier work by Greenlees and colleagues on the impact of body language (Greenlees, Bradley, Holder, & Thelwell, 2005a; Greenlees, Buscombe, Thelwell, Holder, & Rimmer, 2005b; Greenlees, Leyland, Thelwell, & Filby, 2008), results showed that dominant penalty preparations as

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compared to submissive preparations led to more positive impressions and (in turn) lowered participants' confidence in being able to save the anticipated penalty kick. Furthermore, the fact that similar results were observed in a sample of field players and a sample of goalkeepers, supported the notion that implications of body language are general rather than dependent on domain-specific knowledge or expertise (also see Furley & Schweizer, 2014). A second experiment (Furley et al., 2012; Experiment 2), which tested for automatic implicit associations rather than person impressions, showed that this effect is likely attributable to perceivers' expectations. That is, participants associated dominance more easily with positive performance related attributes (positive athlete schema) and submissiveness more easily with negative performance related attributes (negative athlete schema), than the opposite pattern of associations (dominance-negative and submissiveness-positive).

Because classifying other players based on their body language may trigger emotional responses (e.g., anxiety) and expectancies of success that have the potential to influence sports performance (Greenlees, 2007), this prior work (Furley et al., 2012; Greenlees et al., 2005a, 2005b, 2008) suggests that body language may be an influential factor. However, there are several questions that remain. First, it is not clear whether the associations between body language and performance attributes found in Furley, Dicks, and Memmert, (2012); Experiment 2) influence the impression formation process. Second, there is currently no empirical evidence available to indicate whether nonverbal cues of dominance and submissiveness may affect actual performance in observers. In a series of three experiments, the current study aimed to address these issues.

1. Overview of the present research

The current study took the experimental setup of Furley, Dicks, and Memmert, (2012) as basis. In a first experiment, we attempted to replicate Furley et al.'s Experiment 1, using new stimulus material and – in order to examine the robustness of the impression formation effect – aimed to extend their findings by including both the perspective of the goalkeeper and the perspective of the penalty-taker. Subsequently, Experiment 2 explored the underlying processes of impression formation. Specifically, we adjusted Furley et al. (2012) Experiment 2 and tested whether implicit associations about dominant (submissive) penalty-takers moderate the impression formation effect. Finally, in Experiment 3, we conducted a first attempt to directly test the effect of dominant and submissive body language on anticipation performance. Specifically, using a video-based test, we adopted the anticipation paradigm used by Savelsbergh and colleagues (Savelsbergh, Van der Kamp, Williams, & Ward, 2002; Savelsbergh, Williams, Van der Kamp, & Ward, 2005) and examined if goalkeepers anticipation of penalty kick direction was influenced as a function of dominant and submissive pre-performance cues displayed by penalty-takers.

2. Experiment 1

Recognizing the importance of replication (Open Science Collaboration, 2015; Pashler & Wagenmakers, 2012; Simons, 2014; Yong, 2012), Experiment 1 aimed to replicate the findings of Furley, Dicks, and Memmert, (2012). In addition, to enhance the predictive and explanatory range of Furley et al.'s findings, we created new stimulus material, which not only contained video clips displaying the pre-performance behaviors of penalty-takers, but (in separate clips) also the pre-performance behavior of goalkeepers. In line with Furley, Dicks, and Memmert, (2012), we predicted that when target players (penalty-takers and goalkeepers alike) displayed dominant as opposed to submissive body language, this would result in more positive impressions and lower expectancies of success in observers (also see Greenlees et al., 2005a; 2005b; 2008). Because implications of body language have previously been shown to be quite general (i.e., as opposed to being

restricted to specific situations or situation-specific expertise; Furley et al., 2012; Furley & Schweizer, 2014), we anticipated to observe this effect in the penalty-taker as well as in the goalkeeper condition.

3. Methods

3.1. Participants & design

Fifty-Four Radboud University students (40 females, $M_{\text{age}} = 19.56$, $SD_{\text{age}} = 1.75$, range = 18–28), of whom 7 were experienced soccer players ($M = 9.00$ years, $SD = 1.75$, range = 1–13), participated in a video-rating study with a repeated measures design. Participants watched and rated videos of goalkeepers and penalty-takers who prepared for a penalty kick on seven impression formation variables, two quality of the shot/dive variables, and an expectancy of success variable. Targets in these videos showed either a dominant or a submissive body posture.

3.2. Materials & procedure

Development of video materials. Following the procedure of Furley, Dicks, and Memmert, (2012) we created thirty-two video fragments; 2 perspectives (goalkeeper or penalty-taker) x 2 body postures (submissive or dominant) x 2 dive/kick directions (left and right) x 4 target players. The target players were experienced goalkeepers ($n = 4$) and soccer players ($n = 4$). Within perspective, each target player appeared (once) in each condition. In order to standardize the content of the videos all goalkeepers and penalty-takers wore black clothes. Timing and direction of the dive/kick were controlled in the recording of the videos (i.e., target players acted on carefully timed instructions and dived/kicked in direction of a predefined target; only successful trials were included accepted). The goalkeepers' video started with the target entering the goal, turning at the middle of the goal, and taking position (at this moment the video stopped, average length was ~11 s). The penalty-takers' video started with the penalty-taker placing the ball on the penalty mark, walking backwards facing the camera, and initiating the run-up (at this moment the video stopped, average length was ~14 s). As such, neither the actual dive (goalkeeper) nor the actual shot (penalty-taker) were visible from the videos. Fig. 1 shows a video still of a target player in the goalkeeper condition (A/B) and the penalty-taker condition (C/D).

In recording the videos we instructed target players to adopt either a dominant or a submissive body posture. Identical to Furley et al. (2012), the instruction for the dominant body posture was: (a) stand and walk with erect posture with shoulders pulled back and chest pushed out, (b) slightly spread the limbs from the torso to occupy more space, (c) hold the head up with the chin parallel to the ground, (d) gazing directly at the penalty-taker/goalkeeper. The instruction for the submissive body posture was: (a) adopt a slouched posture with head and chin pointing down, (b) limbs touching the torso and minimize occupied space, (c) shoulders hanging to the front, (d) the eyes looking down a lot and only briefly gazing at the penalty-taker/goalkeeper (Furley et al., 2012). Only trials on which target players successfully adopted these postures were accepted.

Experimental task. The computerized experiment (Inquisit, Millisecond Software LLC, Seattle) was executed at Radboud University. Participants watched the 32 videos that were presented in two test-blocks, a goalkeeper and a penalty-taker block. Block order was counterbalanced across participants. The presentation of videos was randomized within each test-block. We added a one-trial practice-block to familiarize participants with the video-rating task before each test-block.

Following Furley, Dicks, and Memmert, (2012), participants had to rate each video on seven impression formation measures: assertive, competitive, experienced, confident, composed, focused, and relaxed. Then, they rated the power of the penalty/dive and the accuracy of

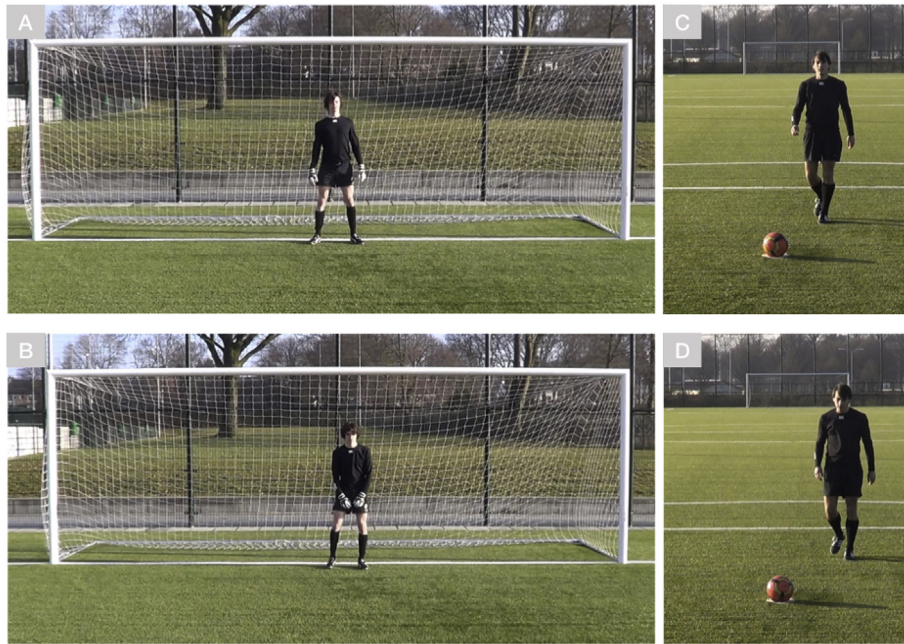


Fig. 1. Image of the target player (video still) in the dominant Goalkeeper condition (A), the submissive Goalkeeper condition (B), the dominant Penalty-taker condition (C), and the submissive Penalty-taker condition.

penalty/dive. Finally, participants rated the expectancy of success (i.e., confidence of the participants that they would score or save at least 1 out of 5, 2 out of 5, 3 out of 5, 4 out of 5, and 5 out of 5 penalties against the displayed target). Each rating was given on a slider ranging from 0 (not at all) to 100 (very much). The mouse cursor was always positioned in the middle of the scale at start. Upon finishing the test-blocks participants were thanked for their participation.

3.3. Data analysis

Following Furley et al. (2012), we averaged participants' ratings per target and body posture combination and computed a single impression formation rating for each participant. Finally, expectancy of success scores were calculated. In order to ascertain more weight to five saves/scores than one save/score, each expectancy rating was multiplied by the amount of saves or scores the participants expected. Therefore, the scores ranged from 0 to 1500. Separate repeated measures ANOVAs were performed with body posture (dominant vs. submissive) and target (goalkeeper vs. penalty-taker) as within-subject factors on (1) the general impression formation, (2) power of the penalty/dive, (3) accuracy of the penalty/dive and computed expectancy of success score.

4. Results

General impression formation. A repeated measures ANOVA on the general impression formation rating revealed that targets displaying dominant body postures were rated more positively than targets displaying submissive body postures, $F(1, 53) = 336.76$, $p < .001$, $\eta_p^2 = 0.86$; Penalty-takers were rated more positive than goalkeepers, $F(1, 53) = 102.20$, $p < .001$, $\eta_p^2 = 0.66$. Furthermore, a two-way interaction between body posture and target was observed, $F(1, 53) = 41.40$, $p < .001$, $\eta_p^2 = 0.44$, showing that the difference in general impression ratings between dominant and submissive targets, was larger for penalty-takers than for goalkeepers (see Table 1).

Accuracy and Power. The repeated measures ANOVAs on the accuracy and power of the penalty/dive also revealed a main effect of body posture (accuracy: $F(1, 53) = 133.74$, $p < .001$, $\eta_p^2 = 0.72$; power: $F(1, 53) = 139.61$, $p < .001$, $\eta_p^2 = 0.73$) and of target (accuracy: $F(1, 53) = 50.17$, $p < .001$, $\eta_p^2 = .49$; power: $F(1, 53) = 133.42$,

$p < .001$, $\eta_p^2 = 0.72$). Penalties/Dives of targets displaying dominant body postures were rated more powerful and accurate than targets displaying submissive body postures; also, penalty-takers were rated more powerful and accurate than goalkeepers. Furthermore, for both measures a two-way interaction between body posture and target was observed, $F(1, 53) = 14.97$, $p < .001$, $\eta_p^2 = 0.22$ (accuracy) and $F(1, 53) = 4.64$, $p = .036$, $\eta_p^2 = 0.08$ (power), showing that the difference in accuracy/power ratings between dominant and submissive targets, was larger for penalty-takers than for goalkeepers (see Table 1).

Expectancy of Success. Finally, a repeated measures ANOVA on the expectancy of success ratings revealed a main effect of body posture, $F(1, 53) = 74.53$, $p < .001$, $\eta_p^2 = 0.58$, and of target, $F(1, 53) = 36.31$, $p < .001$, $\eta_p^2 = 0.41$. Participants expected to be more successful against targets displaying submissive body postures than targets displaying dominant body postures; participants expected to be more successful against goalkeepers than penalty-takers. Furthermore, a two-way interaction between body posture and target was observed, $F(1, 53) = 11.69$, $p = .001$, $\eta_p^2 = 0.18$, showing that the difference in expectancy of success ratings between submissive and dominant targets, was larger when participants watched videos of goalkeepers than penalty-takers (see Table 1).

5. Discussion

By confirming that a target player's display of dominant as opposed to submissive body language, is generally associated with more positive impressions and more negative outcome expectancies in observers, Experiment 1 is consistent with earlier observations by Greenlees et al. (2005a, 2005b, 2008) and fully replicates the findings of Furley, Dicks, and Memmert, (2012); Experiment 1). Moreover, the effect sizes for each dependent variable suggest that body language strongly affected participant's judgments. Although effects appeared to be slightly stronger when the target player was a penalty-taker rather than a goalkeeper, the fact that the same effects were observed across both perspectives and with new stimulus material, underlines the idea that implications of body language are general rather than domain-specific and further enhances the explanatory range of Furley et al.'s findings. In terms of understanding the impression formation process, however, an important question that remains, is whether the player impressions that

Table 1

Overview of the mean scores and standard deviations for each dependent variable in Experiment 1.

Target in video	Penalty-taker		Goalkeeper	
	Dominant	Submissive	Dominant	Submissive
General impression (0–100)	65.13 (7.69)	47.93 (6.25)	60.95 (8.50)	36.36 (8.82)
Accuracy shot/dive (0–100)	61.43 (10.84)	48.58 (10.34)	55.22 (10.88)	35.83 (13.17)
Power shot/dive (0–100)	66.99 (8.96)	52.57 (10.08)	55.89 (10.34)	37.66 (12.64)
Expectancy of success (0–1500)	409.05 (206.66)	512.35 (209.83)	569.44 (247.31)	749.19 (302.21)

were assessed in Experiment 1 are informed by the automatic (implicit) evaluation of player characteristics, or whether they result from more deliberate (explicit) information processing. This was investigated in Experiment 2.

6. Experiment 2

Sometimes deliberative reactions to a person differ from more automatic reactions to that same person, which has led social psychologists to propose two distinct processes of implicit and explicit evaluations (e.g. Smith & DeCoster, 2000; Wilson, Lindsey, & Schooler, 2000). In Furley et al. (2012) study, explicit (Experiment 1) and implicit (Experiment 2) impressions were assessed in two different groups of participants. This approach is limited in gaining understanding of the underlying processes of impression formation as it does not allow to test whether implicit associations influence the explicit impression formation process, or whether there are two independent evaluation systems at work (as is suggested by dual-process accounts of cognition; e.g., Sloman, 1996; see Furley, Schweizer, & Bertrams, 2015, for a review on dual-process accounts in sports). To answer this question, Experiment 2 employed the video-rating task and an implicit association test (IAT; as in Furley et al., 2012, Experiment 2) to assess both explicit impressions and implicit associations of one group of participants. Based on Furley et al. (2012), we expected to replicate the effect of body language on explicit impressions (as in Experiment 1) and to show that participants more strongly associate dominant body language with ‘good soccer players’ and submissive body language with ‘bad soccer players’ than the opposite (dominant-bad and submissive-good; as in Furley et al., 2012, Experiment 2), on an implicit level. Furthermore, we expected that participants’ implicit associations would moderate the relation between body language and explicit impressions. That is, higher IAT-scores would result in more positive explicit general impression ratings and lower expectancies of success in observers of target players that displayed dominant as opposed to submissive body language.

7. Methods

7.1. Participants & design

Sixty-Eight female Radboud University students ($M_{\text{age}} = 22.28$, $SD_{\text{age}} = 3.50$, range = 18–39, no soccer experience measure was included) participated in a video-rating study with a repeated measures design. As in Experiment 1, participants watched videos of target players who displayed dominant or submissive body language while preparing for a penalty kick, and rated each video on seven impression formation variables, two quality of the shot variables, and an expectancy of success variable. After performing the video rating task, participants completed an implicit association test (IAT).

7.2. Materials & procedure

Video-rating task. The materials and procedure of the video-rating task were identical to those of Experiment 1. The only exception was that participants were only presented with the penalty-takers and not with the goalkeepers. Consequently, participants watched one block of

16 videos.

Implicit association test. Upon finishing the video-rating task, participants completed an implicit association test (IAT, Greenwald, McGhee, & Schwartz, 1998). Following the procedure of Furley, Dicks, and Memmert, (2012), Experiment 2), the IAT asked participants to categorize images of dominant vs. submissive penalty-takers and words describing good vs. bad soccer players into different categories (see below). The player images were extracted from the video-rating task and included four images of penalty-takers who displayed dominant body language and four images of penalty-takes who displayed submissive body language. The words describing good vs. bad soccer players were established based on a pilot test (see <https://osf.io/yvmfq/>) and included four words associated with good soccer players (i.e., ‘fit’, ‘certain’, ‘skilled’, ‘fast’) and four words associated with bad soccer players (i.e., ‘weak’, ‘clumsy’, ‘slow’, ‘tired’). The IAT was designed to measure the strength of an individual’s association between the quality of football players and the two body postures.

After two 20-trial practice blocks, in which participants learned to distinguish between images of dominant vs. submissive penalty-takers (practice block 1), and words describing good vs. bad soccer players (practice block 2), participants performed a first test block of 40 trials, in which they categorized words describing good players and images of dominant players by pressing one response key, and words describing bad players and images of submissive players by pressing another response key (i.e., ‘congruent’ block). Following this test block, participants received a practice block, in which only images of dominant vs. submissive players were shown and in which the labels switched sides. That is, the response key that was first used to indicate a dominant player image would now be used to indicate a submissive player image (and vice versa). Subsequently, a second test block of 40 trials was performed, in which all participants categorized words describing good players and images of submissive players by pressing one key, and bad players words and images of dominant players by pressing the other key (i.e., ‘incongruent’ block). Following this procedure, and in keeping with Furley, Dicks, and Memmert, (2012), shorter response latencies in the congruent block than the incongruent block were assumed to reflect stronger automatic associations between good players and dominant body postures, and bad players with submissive body postures, compared to the opposite player-body posture combinations.

In this study, the IAT was used as individual difference measures for predicting participants’ ratings on the video-rating task. In order to minimize between participants differences in IAT-scores due to order effects, blocks as well as words were presented in a fixed, pseudo-random order.

7.3. Data analysis

Before analyzing the data we averaged participants’ responses on the video-rating task, exactly as in Experiment 1. Next, we calculated a measure of associative strength between good/bad players and dominant/submissive body postures (IAT-score) separately for each participant. First, incorrect responses (11%), practice blocks, and the first two trials of all test blocks were excluded from this calculation; as in Greenwald et al. (1998; see also Bijlstra, Holland, Dotsch, Hugenberg, & Wigboldus, 2014). Second, to account for influential outliers, we

trimmed our reaction time data and log transformed each response latency (see Ratcliff, 1993). Response latencies smaller than 300 ms were recoded to 300 ms and those bigger than 3000 ms–3000 ms. Finally, the mean response latency of the congruent block was subtracted from the mean response latency of the incongruent block. Greater IAT-scores reflect stronger good player-dominant body postures and bad player-submissive body postures (relative to the opposite combination of) associations.

First, we analyzed participants' IAT data to see whether we replicated Furley and colleagues findings (2012, Experiment 2). Next, to replicate Experiment 1 and test whether implicit associations underlie these effects, we performed separate repeated measures ANOVA's with body language (dominant vs. submissive) as within-subject factor and participants' IAT-score as continuous measure on (1) the general impression formation, (2) power of the penalty/dive, (3) accuracy of the penalty/dive and computed expectancy of success score. All analyses were performed using log-transformed response latencies, but untransformed latencies are reported for ease of interpretation.

8. Results

Table 2 presents an overview of the mean scores and standard deviations for all dependent variables in Experiment 2.

Implicit association task. In the IAT, participants responded faster during the congruent block ($M = 774$, $SD = 103$) than during the incongruent block ($M = 1172$, $SD = 297$), $F(1, 67) = 178.03$, $p < .001$, $\eta_p^2 = 0.73$. This indicates that, on average, our participants associated dominant body postures more with good players and submissive body postures more with bad players, than submissive body postures with good players and dominant body postures with bad players.

General impression formation. A repeated measures ANOVA on the general impression formation rating revealed that targets displaying dominant body postures were rated more positively than targets displaying submissive body postures, $F(1, 66) = 179.30$, $p < .001$, $\eta_p^2 = 0.73$ (see Table 1). No significant interaction between body posture and IAT-score was observed, $F(1, 66) = 2.05$, $p < .157$, $\eta_p^2 = 0.03$, showing that the difference in general impression ratings of dominant and submissive for penalty-takers was not depending on participants' IAT-score.

Accuracy and Power. The repeated measures ANOVAs on the accuracy and power of the penalty also revealed a main effect of body posture, $F(1, 66) = 80.92$, $p < .001$, $\eta_p^2 = 0.55$ and $F(1, 66) = 98.20$, $p < .001$, $\eta_p^2 = 0.60$ respectively. Penalties of targets displaying dominant body postures were rated more accurate and powerful than targets displaying submissive body postures. Furthermore, for both measures, no significant interaction between body posture and IAT-score was observed, $F(1, 66) = 1.66$, $p = .202$, $\eta_p^2 = 0.03$ (accuracy) and $F(1, 66) = 0.20$, $p = .660$, $\eta_p^2 < 0.01$ (power), showing that the difference in accuracy/power ratings for dominant and submissive penalty-takers did not depend on participants' IAT-score.

Expectancy of Success. Finally, A repeated measures ANOVA on the expectancy of success ratings revealed a main effect of body posture, $F(1, 66) = 74.55$, $p < .001$, $\eta_p^2 = 0.53$. Participants expected to be more successful against targets displaying submissive body postures

than targets displaying dominant body postures. Again, no significant interaction between body posture and IAT-score was observed, $F(1, 66) = 1.34$, $p = .252$, $\eta_p^2 = 0.02$, showing that the difference in expectancy of success ratings of participants against submissive and dominant penalty-takers was not depending on their IAT-score.

9. Discussion

In line with Experiment 1, the results of Experiment 2 confirm that penalty-takers who display dominant body language are generally evaluated more positively (and trigger more negative outcome expectancies) than penalty-takers who display submissive body language (Furley et al., 2012; also see Greenlees et al., 2005a, 2005b, 2008). Congruent with Experiment 1, the observed effect sizes are large. Furthermore, and in keeping with Furley et al. (2012; Experiment 2), results showed that apart from their explicit impressions, observers more strongly associated dominant and submissive body language with 'good' and 'bad' soccer players, compared to the opposite associations. Interestingly, the complete absence of significant interaction effects (all p 's > 0.20 , all η_p^2 's < 0.03) suggests that participants' implicit associations (as measured with the IAT) did not affect the formation of their explicit impressions (as measured with the video-rating task). The small effect sizes suggest that these null-findings are not due to a lack of power. This finding indicates that even though implicit associations and explicit impressions of target players may align conceptually (as in the current experiment), we do not find evidence for these measures to affect each other in the current person perception processes.

10. Experiment 3

With the impact of body language of dominance and submissiveness on impression formation and outcome expectancies being firmly established (Experiment 1 and 2, Furley et al., 2012; also see Greenlees et al., 2005a, 2005b, 2008), an important question that remains unanswered is whether there may also be consequences for behavior and performance. Previous research on body language and impression formation in sport has mostly neglected to test such consequences (see Furley et al., 2012 for an exception) and has only speculated about how perceptions of dominance might influence further behavior. However, dominant body language has indeed been shown to affect behavior in social interaction (Tiedens & Fragale, 2003). Prior research and theorizing has mainly assumed an "indirect" path of how body language might affect performance in penalty situations: based on the body language of penalty-takers, goalkeepers form impressions of their opponent which may trigger emotional responses (e.g., anxiety or confidence) and expectancies of success that have the potential to influence performance (Feltz, Short, & Sullivan, 2008; Greenlees, 2007). This reasoning has the potential to explain performance effects in real-world penalty shootouts, but is difficult to test within the laboratory-based experimental scenario adopted in Experiments 1 and 2.

Against this background, and building on the results of Experiments 1 and 2, Experiment 3 examined whether or not observing nonverbal cues of dominance and submissiveness in opponents may significantly impact anticipation performance in a soccer penalty situation. In penalty situations, goalkeepers must anticipate kick direction based on advance perceptual information, i.e. bodily cues from the penalty-takers, in order to be on time with their response and successfully stop the penalty (e.g., Savelsbergh, Williams, van der Kamp, & Ward, 2002; 2005). In Experiment 3, we integrated Furley et al. (2012) video rating task and the anticipation paradigm used by Savelsbergh and colleagues (Savelsbergh et al., 2002; 2005), and tested if anticipation of kick direction (i.e., left or right) was influenced by the penalty-takers' body language. Furthermore, in case of significant performance effects, mediation analyses were performed to examine if the effect of body language is likely to be mediated (in part) by player impressions. Based on the evolutionary function of dominance cues (Mazur, 2005; Rule

Table 2

Overview of the mean scores and standard deviations for each dependent variable in Experiment 2.

Target in video	Penalty-taker	
	Dominant	Submissive
General impression (0–100)	62.43 (7.46)	48.83 (6.87)
Accuracy shot (0–100)	60.42 (9.30)	50.87 (8.60)
Power shot (0–100)	64.03 (8.69)	52.26 (8.69)
Expectancy of success (0–1500)	334.02 (210.24)	432.57 (241.11)

et al., 2012; Zebrowitz & Collins, 1997; Furley and Schweizer, 2016), as well as the speculations put forth in previous investigations of body language in sports (e.g., Furley et al., 2012), we expected that participants' anticipation performance would be worse against dominant than submissive penalty-takers. Moreover, we expected that this effect would be (partially) mediated by player impressions, i.e., the better participants think that penalty-takers are, the worse they perform.

11. Methods

The study design, sample size, materials, hypotheses, and statistical analyses of Experiment 3 were preregistered on the Open Science Framework and can be accessed at: <https://osf.io/s32un/>.

11.1. Participants & design

We based our sample size on an a priori power analysis using G*Power (Faul, Erdfelder, Lang, & Buchner, 2007), such that there was an 80% chance to detect a small to medium sized effect at an alpha level of 0.05. This power calculation showed that we needed at least 60 participants for the current study (see Preregistration). A total of seventy-two Radboud University students (45 females, $M_{\text{age}} = 21.99$, $SD_{\text{age}} = 3.11$, range = 18–33) of whom 14 were experienced soccer players ($M = 9.79$ years, $SD = 4.95$, range = 1–17), participated in a video-rating study with a repeated measures design. As in Experiments 1 and 2, participants watched and rated videos of penalty-takers taking a dominant or submissive body posture while preparing for a penalty kick. More importantly, Experiment 3 required participants to make anticipatory judgments regarding the side of the goal to which each ball was shot (left or right; Savelsbergh et al., 2002; 2005).

11.2. Materials & procedure

For Experiment 3 we created a new set of video stimuli, which was developed using the same procedure as those of Experiment 1 and 2, but (for each penalty-taker) included shots targeted to the left and right side of the goal. Videos of 10 penalty-takers were recorded indoors, at the University Sports Centre. In total this led to 40 videos: 10 penalty-takers showing 2 body postures (submissive and dominant) $\times$ 2 shot directions (left or right). Timing and direction of the penalty kicks were controlled in the recording of the videos (i.e., target players acted on carefully timed instructions and kicked with consistent force in the direction of a predefined target). No deceptive strategies were used and only successful trials (i.e., on which target players adhered to instructions) were accepted. Post-hoc examination of the raw video material (frame-by-frame analysis of a representative subset of trials [i.e.50%; including all target players]) indicated that speed of the penalty kicks did not differ significantly between conditions ($F < 1$, $p = .68$). The footage of one penalty-taker (i.e., 4 videos) was used to familiarize participants with the experimental task. Footage of the remaining 9 penalty-takers, was presented in random order, in a 36-trial test-block.

Each video was cut at 240 ms (6 frames) before foot-ball contact (cf. Savelsbergh et al., 2002; 2005), using Adobe Premiere Pro CC (Version 9.0.2; Adobe Systems Incorporated, 2015). The 240 ms cut-off point was based on an initial pilot study, which – using participants without substantial soccer experience – resulted in an average anticipation performance of 69% correct responses. This performance level was deemed ideal, as it prevents the study from having floor or ceiling effects and thus leaving enough room for body language to have a potential effect. Similar to Savelsbergh et al. (2002; 2005), using the 240 ms cut-off point, participants could see the entirety of the preparation and run-up until the point where the standing leg was planted next to the ball and the kicking leg was about to swing forward.

Immediately after watching each video (i.e., as soon as possible), participants were asked to judge the side of the goal to which the ball was shot by pressing a response key that corresponded with their

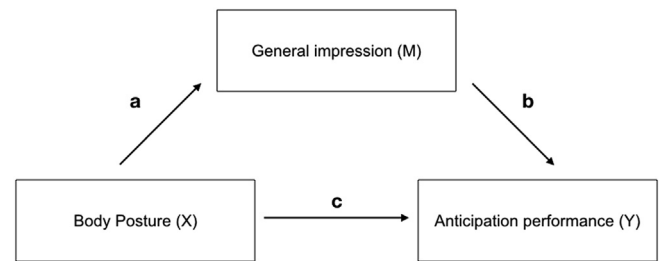


Fig. 2. Anticipated mediation model in Experiment 3.

judgment (i.e., left or right). Next, participants rated the penalty-taker on three impression formation items (i.e., *noncompetitive* – *competitive*; *novice* – *experienced*; and *unconfident* – *confident*; 1–100 scales). Since the impression formation items correlated substantially in Experiment 1 and 2, we reduced the number of items in Experiment 3 in order to save time.

11.3. Data analysis

Before analyzing the data, we averaged participants' impression ratings and anticipation performance per body posture combination. To test for mediation, four steps should be established (see Fig. 2, Baron & Kenny, 1986). First, in order to see whether we replicated Experiments 1 and 2 with our new stimulus material, we ran an ANOVA with body posture as within-subjects variable and the potential mediator general impression as dependent variable (path a). Next, we analyzed participants' anticipation performance using an ANOVA with body posture as within-subjects variable and percentage of correct judgements as dependent variable (path c). Finally, to explore whether participants' anticipation performance could be explained by participants' impression of penalty-takers, we applied Generalized Estimating Equations (path b). To maximize the power of this step, we predicted participants' anticipation performance from participant's general impression ratings on a trial by trial basis (i.e., instead of averaging over trials). Please note that penalty-takers' body posture was coded within each trial. Only in case that these three steps revealed significant relations, evidence for (partial) mediation was tested by looking at path c', which is the effect of body posture (X) on anticipation performance (Y), while controlling for participants' impression of penalty-takers (M). Based on the previously mentioned pilot, we decided to exclude difference scores on any of the variables that were 3SD above or under the mean.

12. Results

Table 3 presents an overview of the means and standard deviations for each dependent variable in Experiment 3.

General Impression Formation. The ANOVA for impression formation revealed that when penalty-takers displayed a dominant body posture, they were rated more positively than when they displayed a submissive body posture, $F(1, 71) = 194.53$, $p < .001$, $\eta_p^2 = 0.73$.

Anticipation Performance. The ANOVA for anticipation performance (percentage of correct judgements) revealed that participants performed significantly better on trials in which penalty-takers displayed a

Table 3

Overview of the mean scores and standard deviations for each dependent variable in Experiment 3.

Target in video	Penalty-taker	
	Dominant	Submissive
General impression (0–100)	58.75 (6.89)	43.15 (9.14)
Anticipation Performance (% correct)	70% (14%)	74% (17%)

submissive body posture than on trials in which penalty-takers displayed a dominant body posture, $F(1, 71) = 4.68, p = .034, \eta_p^2 = 0.06$.

Impact of General Impression Formation on Anticipation Performance. The Generalized Estimating Equations analysis that was performed on participants' (trial-by-trial) impression ratings and anticipatory judgements, revealed no significant effects. A non-significant effect of impression formation on performance was observed (GEE: Wald Chi square = 1.53, $df = 1, p = .216$). Since no relation was found here, no (partial) mediation was present in the current data and step 4 of the mediation analysis was not performed.

13. Discussion

Using new stimulus material, the results of Experiment 3 replicated Experiments 1 and 2 by showing once again that penalty-takers who display dominant body language are generally regarded more positively (i.e., as 'better players') than penalty-takers who display submissive body language (Furley et al., 2012; also see Greenlees et al., 2005a; 2005b; 2008). In addition, Experiment 3 provides initial evidence that the body language displayed by penalty-takers during the preparation of their kick, may significantly influence goalkeepers' anticipation performance, as participants performed significantly less accurate against dominant players than against submissive players. Following Cohen (Cohen, 1962), the observed effect size suggests a medium sized effect. Finally, regarding the potential mechanisms underlying this effect, we did not find evidence for the idea that this could be attributed to participants' impression of target players (cf. Furley et al., 2012; Greenlees, 2007).

14. General discussion

The goal of the present research was to further understanding of how pre-performance body language affects impression formation and performance in soccer penalty kicks. In each Experiment, we were able to replicate the results of Furley, Dicks, and Memmert, (2012) and further show that the results held up with different stimulus material displaying both penalty-takers and goalkeepers (Experiment 1). Direct replications are a research necessity and should not be regarded a trivial research step considering the increasing calls for replication in the psychological literature (Open Science Collaboration, 2015; Pashler & Wagenmakers, 2012; Simons, 2014; Yong, 2012). In Experiment 2, the data suggest that impression formation based on dominant and submissive body language in sports may rely on two distinct processes—one automatic, associative process and one rule-based, explicit process (e.g., Smith & DeCoster, 2000)—as explicit and implicit measures were not related to one another. However, we should be careful drawing too strong conclusions based on this null finding, especially because the conceptual correspondence between our implicit and explicit measures may be quite low (for more information on the relation between implicit and explicit measures see Gawronski, 2019). Finally, and most importantly, Experiment 3 showed that participants' anticipation performance declined when anticipating kick direction of dominant as opposed to submissive penalty-takers. However, no evidence was found for mediation by players' impressions, an indirect effect, in this analysis.

Penalty situations are a suitable situation to study the nonverbal communication of dominance and submission due to their intense one-on-one nature with both the penalty-taker and goalkeeper facing each other when preparing their respective performances. Of applied importance, body language is under both conscious, deliberate control and under unconscious, autonomous control. This means that body language can unintentionally leak certain internal information, but a player's body language can also be modified for self-presentational goals and favorable social outcomes (DePaulo, 1992), as perceivers rapidly and spontaneously make judgments and form impressions based on nonverbal cues (Argyle, 1972; Bijlstra, Holland, Dotsch, &

Wigboldus, 2019). Research suggests that people can deliberately use their body language for self-presentational purposes, although situational and dispositional constraints influence the degree to which this is successful (see DePaulo, 1992, for a review). In consideration of the findings from Experiment 3, it seems advisable to assist athletes in the development of positive self-presentation techniques, especially in situations of high stress in which people tend to leak information they would rather hide (Furley & Schweizer, 2017). Specifically, penalty-takers likely would benefit from expressing dominance as this may result in lowered goalkeepers' anticipation performance.

Why are goalkeepers affected by penalty-taker's body language? Contrary to our predictions, no evidence was observed for an 'indirect effect' in which participants' impression of penalty-takers explained the effect of body language on anticipation behavior. Alternatively, it could be that perceptions of opponents' body language may have influenced goalkeepers more directly. That is, there is reason to believe that visual attentional processes may have influenced the information pickup which is required to accurately perform the anticipatory judgements. Regarding such a 'direct effect', Furley et al. (2017) have provided evidence that athletes who are perceived as more dominant are attended to in a different manner (i.e., elicit an early attentional orienting response in observers) compared to submissive athletes. This observation is supported by the ecological theory of person perception (McArthur & Baron, 1983; Zebrowitz & Collins, 1997), which states that our cognitive system has become particularly attuned to perceptual cues that are of general adaptive relevance, such as status or dominance cues (see Rule et al., 2012; Schmid Mast & Hall, 2004). Indeed, in the soccer penalty situation adopted in the current experiments, the tendency of individuals' attention being drawn to dominant cues (de Waal, 2007; Zebrowitz & Collins, 1997) might cause goalkeepers to attend more to penalty-takers' upper body (i.e., where dominance cues are typically placed; Carney, Hall, & Smith LeBeau, 2005) and miss out on task-relevant information that helps them to effectively anticipate the direction of the kick (i.e., which tends to be located in the penalty-takers' lower body; Savelsbergh et al., 2002; 2005).

The present research has some notable strengths, such as that it includes replications, being well powered and (partly) preregistered. Most importantly, with Experiment 3 we initiated the investigation of the relation between impression formation and actual behavior. However, the studies also have some limitations. First, while we did not find evidence for the effect of body language on anticipated performance to be 'indirect', or mediated (in part) by player impressions, we cannot conclude that the effect is 'direct'. In order to test this, a study is needed in which participants' visual attention is measured while watching videos using an eye-tracking device (see for example Savelsbergh, van der Kamp, Williams, & Ward, 2005). If participants' visual attention is indeed affected by penalty-takers' body language, and therefore is less optimal for inferring relevant information, then anticipation performance may be influenced consequently.

Second, the generalizability of our results, both to other participant samples and test-situations, remains untested. Although we fully replicated previous studies, and the effect of body language seems to be general rather than domain-specific (i.e., independent of target type - see Experiment 1 - or participants' soccer experience - see Furley et al., 2012), we cannot conclude that anticipation performance of expert goalkeepers is also affected by penalty-takers' body language. Potentially, expert goalkeepers have learned to focus their attention on the most diagnostic information and therefore are less strongly affected by body language of penalty-takers. Future studies could systematically investigate whether the effects found in Experiment 3 generalize to expert goalkeepers as well. Furthermore, the classic anticipation paradigm that was employed in the current study is well established in the literature and effectively distinguishes between novices and experts (e.g., Savelsbergh et al., 2002; 2005), but reduces goalkeepers' anticipation performance to a one-dimensional judgment of 'left' or 'right', thereby ignoring other aspects such as height of the kick, force of the

kick and response planning. Similarly, it should be noted that the current paradigm employed video stimuli instead of testing participants' impression formation and anticipation performance in a 'real life' penalty situation (cf. Dicks, Button, & Davis, 2010). Whether our results generalize to actual behavior outside of the lab is an interesting and important avenue for future studies.

In sum, the current research provides strong evidence for the effect of body language on impression formation processes. Furthermore, by concurrently assessing anticipation performance, we found initial evidence that body language may indeed affect goalkeepers' behavior. Future work could further investigate what processes underlie the observed behavioral effects. We argue that a likely candidate for this can be found in direct effects, such as visual attention of goalkeepers.

Declaration of competing interest

Grants and funding: None.

Our research has been conducted conform the ethical standards of the American Psychological Association. At the time of data collection ethical approval was not required for non-invasive research at the institute where the research was conducted.

The current submission does not overlap with any other published, in press, or in preparation articles, books, or proceedings. Our research is not under consideration elsewhere.

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